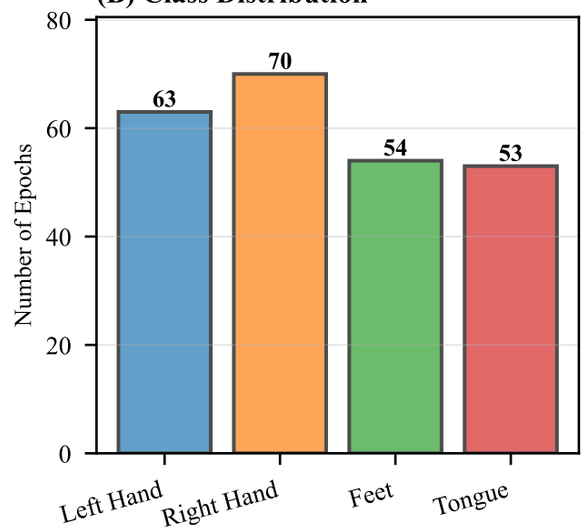
Figure 6: Phase 2 Summary Dashboard - Final Data Quality Assessment

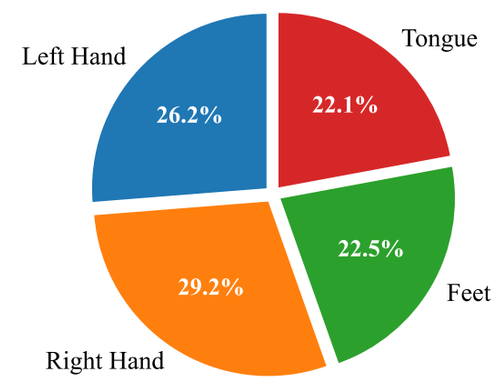
(A) Data Dimensions

```
Final Preprocessed Data

Shape: (240, 25, 751)

• Epochs: 240
• Channels: 25
• Samples/epoch: 751
• Duration: 3.0 seconds
• Sampling rate: 250 Hz

Status: OK Ready for Phase 3
```

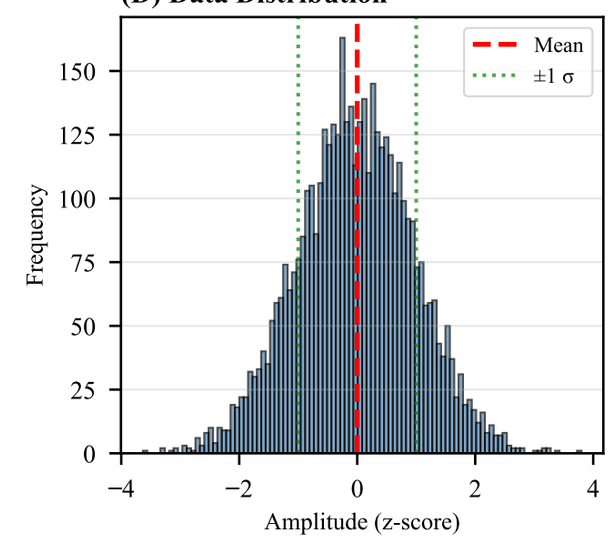
(B) Class Distribution



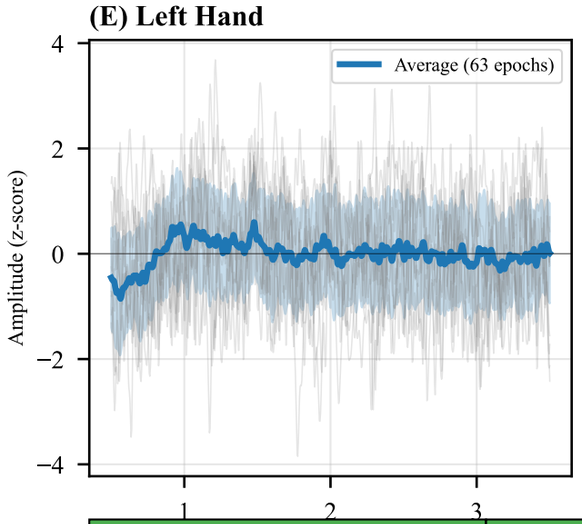
(C) Class Balance



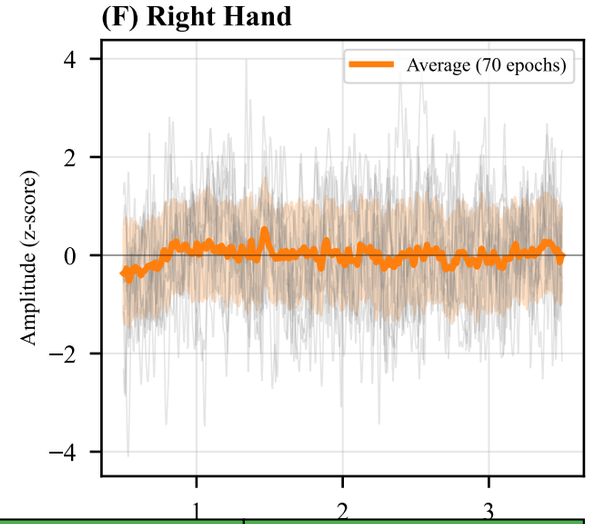
(D) Data Distribution



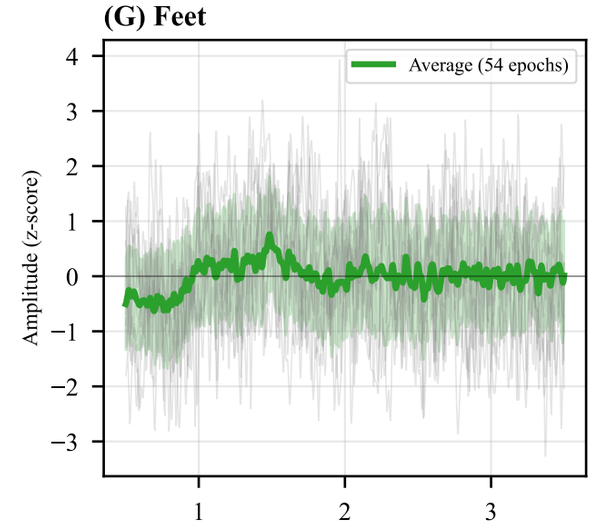
(E) Left Hand



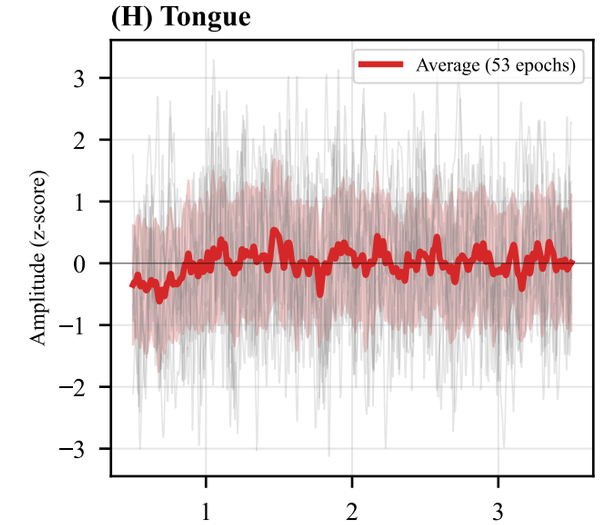
(F) Right Hand



(G) Feet



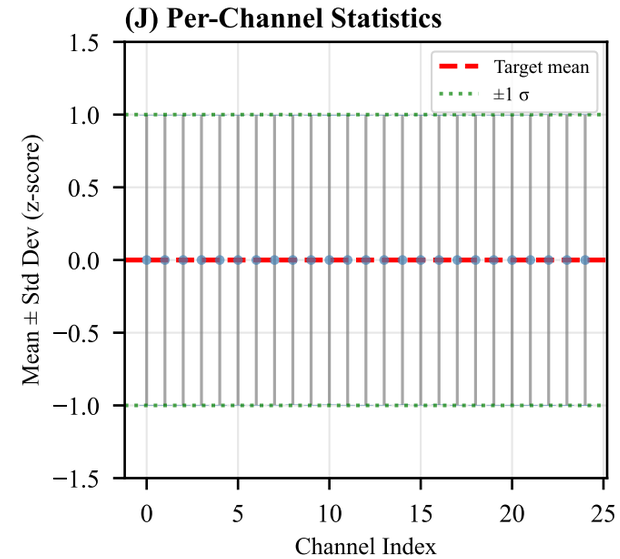
(H) Tongue



(I) Statistical Summary

Metric	Value	Status
Total Epochs	240	OK
Channels	25	OK
Samples per Epoch	751	OK
Mean	-0.000000	OK (target: 0)
Std Dev	0.997445	OK (target: 1)
Min Value	-5.34	OK
Max Value	5.24	OK
Class Balance	±28.3%	OK

(J) Per-Channel Statistics



(K) Pipeline Status

```
Preprocessing Pipeline
──────────────────────
OK Raw data loaded (GDF)
OK Band-pass filter (0.5-40 Hz)
OK Notch filter (50 Hz)
OK ICA artifact removal (2 comp.)
OK Epoch extraction (3.0 sec)
OK Artifact rejection (3 epochs)
OK Z-score normalization

Status: Complete OK

Next: Phase 3
  → Image Transformation
  → GAF, MTF, RP, etc.
```